

Chapter 53 — Surgery of the Ureter in Children: Ureteropelvic Junction, Megaureter and Vesicoureteral Reflux

URETEROPELVIC JUNCTION OBSTRUCTION

Definition

- **Paediatric hydronephrosis** is defined by an **anteroposterior (AP) renal pelvic diameter of 12 mm**; it differs from adult hydronephrosis and is **not synonymous with obstruction**.
- **Koff's definition** is the only gold-standard prognosticator of **ureteropelvic junction (UPJ) obstruction (UPJO)**: obstruction is “**any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage.**”
- The definition is **retrospective**: the indication for surgery is often established only after renal function has been irrevocably lost.
- Guidelines try to get around this, but with **imprecise diagnostic tools, surgical indications remain debatable**.

Concept — Obstruction in the child is defined by its **consequence, not its appearance**. Dilatation is only a surrogate, so every indication for surgery in this chapter is an attempt to predict harm before it happens.

Clinical Presentation

- **Antenatal ultrasound at about 20 weeks' gestation: urogenital anomalies are the dominant malformations**, and among them **hydronephrosis or hydroureteronephrosis is the commonest**.
- **Postnatal diagnosis is by ultrasound**, either incidentally or during workup of **urinary tract infection (UTI)**. Febrile UTI usually responds to antibiotics; only a **minority need nephrostomy** when there is coexisting dilatation.

	Intrinsic UPJO	Extrinsic UPJO
Typical age	Newborns and infants (commonest cause at this age)	Older children and adolescents
Lesion	Narrowed UPJ segment with interrupted development of circular muscle fibres → functional discontinuity of contraction → poor pelvic emptying	Accessory lower-pole vessel crossing anteriorly; the ureter kinks over it
Presentation	Antenatal or incidental dilatation	Intermittent loin pain, nausea and vomiting

- It is **unclear whether an intrinsic factor coexists** with the crossing vessel in symptomatic cases.
- **Epidemiology: male predominance**, especially in infants and toddlers; **left-sided predominance; bilateral in 10–40%**; possible **familial clustering**.

Figure legend, Fig. 53.2 — When the ureter is mobilised off a crossing vessel, there is often no intrinsic narrowing. Even so, **dismembering the ureter and transposing it anterior to the vessel prevents recurrence**.

Secondary Ureteropelvic Junction Obstruction

- **High-grade vesicoureteral reflux (VUR)** with a **tortuous dilated ureter** can kink or narrow the UPJ. Whether it resolves on its own or needs correction of both lesions is unclear.

- Endoscopic injection is unreliable for high-grade VUR, which pushes towards open reimplantation. **Simultaneous reimplantation plus dismembered pyeloplasty is discouraged** because it risks devascularising the ureter from both ends.
- A **subadventitial dismemberment of the UPJ** has been described to protect the blood supply. Despite weak evidence, the advisable course is **conservative management with a two-stage operation**.

Lower Pole Ureteropelvic Junction Obstruction

- **Lower-moiety UPJO in a duplex kidney is uncommon.**
- **Incomplete duplication** → **ureteropyelostomy of the upper-pole ureter to the lower-pole pelvis**, side-to-side or end-to-side.
- **Complete duplication** → **standard dismembered pyeloplasty**, guided by the function of each moiety and the length of the stenosis.
- A **non- or minimally functioning moiety ($\leq 5\text{--}10\%$ of total function on nuclear scan)** → **partial nephrectomy**, not repair.

Figure legend, Fig. 53.3 — In a duplex kidney with lower-UPJ stenosis, a **V-plasty** can resolve the obstruction.

Associated Anomalies

- **Up to 50% of affected infants have another urological abnormality.**
- The commonest is **contralateral UPJO (10–40%)**, followed by **renal dysplasia** → **multicystic dysplastic kidney** → **renal agenesis**.
- **Horseshoe kidney** is rare but associated, through either a true stricture or accessory vessels. **VATER syndrome** also has some association.
- **VUR occurs in up to 40% of children with UPJO**, with a **high rate of spontaneous resolution**.

Surgical Indication

- **About one-third need surgery; most cases resolve with time.**
- **Accepted indications:** increasing AP diameter on ultrasound · **low or falling differential renal function (DRF)** · **breakthrough infection on prophylaxis** · **symptoms** (pain in older children).
- In infants, UPJO can cause **failure to thrive, feeding difficulty and constipation**.
- **Haematuria** comes from rupture of mucosal vessels in the dilated pelvis. UPJO is a **rare cause of hypertension**, through functional deterioration and **renin–angiotensin activation**.

Surgical Repair

- **In infants, open surgery is preferred**, because the incision is hardly larger than laparoscopic or robotic access requires. The minimally invasive route is feasible and gives **less pain and better cosmesis**.
- **Most paediatric urologists prefer the extraperitoneal approach when operating open.** For minimally invasive surgery, the transperitoneal versus retroperitoneal debate continues, with few proper comparative or randomised studies.

	Transperitoneal (laparoscopic or robotic)	Retroperitoneal (retroperitoneoscopic)
Argument for	Easier; familiar anatomical landmarks aid orientation	Avoids intra-abdominal organ injury; leakage stays extraperitoneal; shorter stay; mirrors the open route
Argument against	Risk to bowel; urinary ascites → ileus; longer stay	Longer learning curve; confined working space; difficult orientation

	Transperitoneal (laparoscopic or robotic)	Retroperitoneal (retroperitoneoscopic)
Randomised evidence	The one prospective randomised study favoured the retroperitoneal route : shorter operating time and stay, faster return of bowel function and earlier feeding. Another comparison found a longer operating time retroperitoneally. Overall, no significant outcome difference .	
Not suitable for	—	Horseshoe kidney (pelvis approached from behind) and retrocaval ureter (at least robotically)
Ideal for	—	Robotic repair of lower-moiety UPJO in a duplex system

- **Ease alone does not justify an access route.** Organ injury, leakage and length of stay tip the balance towards the retroperitoneum, and the authors advocate **retroperitoneoscopic pyeloplasty** despite its learning curve.

Dismembered Pyeloplasty

- **Anderson-Hynes dismembered pyeloplasty** is the **gold standard** whatever the access route. It was originally devised for a **retrocaval ureter**, and its **success rate is about 95%**.
- **Why dismember**: a **crossing vessel is preserved** (the ureter is transposed anterior to it), the **adynamic segment is excised** and **redundant pelvis is reduced**.
- **Long dysplastic segment** → **full renal mobilisation** bridges several centimetres. A **horizontal incision in the lower pelvis** adds length for a tension-free anastomosis (“bypass” configuration, Fig. 53.5).

Sequence: open Gerota fascia → dissect the anterior or posterior UPJ, according to access → **free the lower pole so an accessory vessel is not missed**, especially retroperitoneoscopically → use **bipolar** diathermy sparingly to protect the pelvic and ureteric blood supply → stay sutures in the pelvis above the planned division and in the ureter at the stenosis → **decompress a tense pelvis with a 21-gauge needle** to avoid over-reduction → dismember and reduce the pelvis if indicated → **spatulate laterally** well beyond the stenosis into healthy ureter → **leave the stenotic segment attached as a handle** (“no-touch”, less mucosal oedema) → anastomose with **6-0 or 5-0 absorbable suture**, interrupted or continuous, **tension-free at the crotch of the “V”** → avoid excess stay-suture tension, which kinks the ureter once released → **excise the stenotic segment just before completion** → **irrigate the pelvis** to clear clot → return the kidney and cover the anastomosis with perinephric fat. **A Penrose drain is usually not needed.**

- **Stenting** (externalised nephro-ureteric versus internal double-J) is a matter of surgeon preference. In infants, an **externalised stent reduces complications and length of stay compared with no stent**.
- **Removing a double-J usually needs another anaesthetic.** An **extraction string** avoids this but **raises the risk of febrile UTI**, especially in girls with previous UTI. Complication rates for pigtail and externalised stents do not differ.
- **Stent duration of 1–4 weeks does not affect outcome.**

Figure legends, Figs 53.4–53.6 — Traction sutures go on the medial and lateral dependent pelvis and the lateral proximal ureter to keep orientation. **The apex of the lateral spatulation goes to the inferior border of the pelvis**, and the medial ureter to the superior edge. The anastomosis is **full-thickness and watertight**. In reduction pyeloplasty, the redundant pelvis is excised between traction sutures and the cephalad pelvis is closed down to the dependent part. In the **“bypass” pyeloplasty**, a side-to-side anastomosis is made **without transecting the UPJ**, placing the new UPJ **within 1 cm of the lower-pole parenchyma** and keeping the ureter anchored (“no-touch”).

Nondismembered Pyeloplasty

NONDISMEMBERED AND ALTERNATIVE PYELOPLASTY TECHNIQUES VERSUS THE GOLD STANDARD

Technique	Best indication	Key point
Anderson-Hynes (dismembered)	All UPJO; crossing vessel	Gold standard; about 95% success
Heineke-Mikulicz (nondismembered)	—	Inferior to dismembered pyeloplasty in a laparoscopic comparative series
Foley Y-V-plasty	High ureteric insertion, no accessory vessel	Randomised against Anderson-Hynes: numerically lower success, not statistically significant
Vertical flap	Dependent UPJ at the medial margin of a large box-shaped extrarenal pelvis	Base horizontal on the dependent pelvis between UPJ and parenchyma
Spiral flap	Long proximal stricture with an already dependent UPJ	Base oblique, lateral to the UPJ; length-to-width ratio $\leq 3:1$ preserves flap vascularity
"Bypass" pyeloplasty	Possible alternative	No review available

- Flaps are reserved for **severe hydronephrosis with a long stenotic segment** where a tension-free primary anastomosis is impossible. In both flaps, **apex position sets flap length**, which must match the ureteric gap.

Figure legend, Fig. 53.7 (Foley Y-V-plasty) — The **base of the V** lies on the dependent medial pelvis and its **apex at the UPJ**. The stem of the Y runs down the lateral ureter into normal calibre. The flap apex is brought to the lowest point of the ureterotomy, and the posterior then the anterior walls are closed.

Concept — Every pyeloplasty variant answers the same two questions: **is there a crossing vessel, and is the gap too long for a tension-free anastomosis?** Dismembering handles the vessel; flaps and renal mobilisation handle the gap.

Surgical Approach to the Ureteropelvic Junction Obstruction

	Posterior lumbotomy	Flank approach
Patient	Infants and toddlers only (dorsal musculature thickens with age)	Any age
Position	Prone	Supine, index side raised on a gel roll
Access	Vertical incision in the lumbodorsal fascia, retracted laterally → Gerota → direct access to the UPJ	Small transverse incision at the tip of the 11th–12th rib → muscle-splitting → peritoneum swept medially → Gerota opened vertically

Figure legend, Fig. 53.10 — For left posterior lumbotomy, the skin incision runs from the **angle of the 12th rib to the iliac crest**, meeting the crest **one-third of the way from the anterior superior iliac spine to the spinous processes**.

Minimally Invasive Techniques

Endoscopic approach.

- Antegrade and retrograde endopyelotomy became popular in the late 1980s–1990s with **variable results**. **Balloon dilatation under 18 years performs poorly long-term**. Endopyelotomy does **better for secondary (scar) UPJO** than for primary UPJO.
- Antegrade endopyelotomy is limited to **older children and adolescents with moderate hydronephrosis**. It needs **preoperative vessel screening** by computed tomography (CT), magnetic resonance imaging (MRI) or endoluminal ultrasound, plus **fluoroscopy and its radiation exposure**, so it is unattractive in children.
- **Its niche is failed primary pyeloplasty**, where the anatomy and the absence of a vessel are already known → **double-J for 6 weeks**. Even so, **redo pyeloplasty remains the best choice in most cases**.

Figure legend, Fig. 53.11 — Persistent UPJ obstruction 2 months after pyeloplasty, after a nephrostomy for postoperative *Staphylococcus aureus* UTI, treated with an **Acucise cutting balloon**. The waist on the balloon marks the narrowing, and the **cutting wire is positioned laterally**.

Laparoscopic pyeloplasty.

- First reported in **1993**; the transabdominal paediatric series and the **retroperitoneal approach in infants and children** followed. It is now the **method of choice at many centres**.
- Meta-analysis: **laparoscopic outcome is comparable to open**, with robotics an attractive alternative.
- **Transperitoneal or retroperitoneal**: the retroperitoneal route is theoretically safer and is the open route of choice, but comparisons show **no significant difference in outcome or complications** (see the access table above).

Robot-assisted pyeloplasty.

- Paediatric reports began in the early 2000s. As with laparoscopy, the **transperitoneal route is more widely used**, because the retroperitoneal space in children is confined, orientation is difficult and the instruments are relatively large.
- **Shorter, more efficient learning curve than laparoscopy**, owing to three-dimensional (3D) vision and instrument dexterity.
- Compared with open surgery: **comparable outcome, with faster resolution of hydronephrosis** in the robotic group.

Technique of laparoscopic or robotic-assisted pyeloplasty.

	Transperitoneal	Retroperitoneal
Position	45–60° flank	80–90° flank ; gel roll under the contralateral iliac crest ; upper leg straight, lower leg bent; table NOT flexed (flexion reduces the AP depth of the retroperitoneum); positioning unsuitable for adults
Access	Camera port by open Hasson technique ; instrument ports under vision, more medial for a large pelvis or a small infant; Da Vinci Xi ports in the midline	Muscle-splitting at the camera site → open lumbodorsal fascia → balloon-develop a 200–300 mL working space (home-made: 12-Fr catheter with a glove finger ; commercial balloons are too large for small children)
Route to the UPJ	Left side: transmesocolic , which is shorter, less morbid and gives a shorter stay than retrocolic, with equal outcome. If the pelvis is not visible, follow the ureter up to it while watching for an accessory vessel	Surgeon dorsal for laparoscopy (small child may lie diagonally); robot docked cranially , with a fixed camera on older systems. Extra 5-mm assistant port in the iliac fossa for sutures and suction
Traction	Stay suture on the UPJ	Pelvic stay suture through the abdominal wall or fixed to psoas

Figure legends, Figs 53.12–53.14 — Port placement for the transabdominal and retroperitoneal (standard and robotic) approaches, and retroperitoneoscopic positioning, as summarised in the table.

Stones and congenital hydronephrosis. Pelvic or renal stones found at pyeloplasty → **removal with a flexible ureteroscope** (or flexible cystoscope) passed through a laparoscopic or robotic port.

Vascular hitch.

- Hellström and Vahlquist's **vascular hitch** (1951) is regaining favour laparoscopically because it is **easier than pyeloplasty**. The vessel is freed from the UPJ, **moved cranially on the pelvis and embedded in pelvic tissue with non-absorbable sutures**.
- **Only for an anterior lower-pole crossing vessel with NO intrinsic UPJ obstruction**. Selection criteria: moderate hydronephrosis, **no calyceal dilatation**, well-preserved cortex, poor drainage with preserved split function, lower-pole crossing vessel. **At operation, a normal ureter and UPJ with visible peristalsis**.
- Long-term data are limited but **comparable to dismembered pyeloplasty**. The feared **hypertension from embedding the vessel has not been observed**.

Stenting

- **Most surgeons stent** the UPJ, open or minimally invasive. It lowers the risk of **leakage and urinoma** and, transperitoneally, of **ileus from urinary ascites**.
- **Stentless repair** demands meticulous tissue handling to prevent oedema, obstruction and leak. Anderson and Hynes themselves held splinting to be “against all the principles of plastic procedure.”
- **Antegrade double-J** can be hard to pass through the narrow infant ureterovesical junction (UVJ). Bladder position is confirmed with **methylene blue instilled into the bladder**.
- **Retrograde stent** placed just before surgery makes the anastomosis harder. If placed days to weeks before, as an emergency, **UPJ inflammation** makes repair more difficult. **Emergency diversion close to definitive surgery should therefore be a nephrostomy**, which inflames the UPJ less.
- **Extraction-string or magnetic retrograde double-J** avoids the second anaesthetic.

STENT OPTIONS AFTER PYELOPLASTY

Stent	Removal	Bladder catheter	Particular point
Internal double-J	Second anaesthetic (unless string or magnet)	Justified for 24–48 h: the stent causes reflux, transmitting bladder pressure to the pelvis	String: more febrile UTI
Externalised nephro-ureteric (“Blue stent”)	Pulled at 7–14 days without anaesthesia	Unnecessary: the lower end is cut short and does not cross the UVJ	Exits through a middle calyx or the upper anastomosis and the flank. Open for 12–48 h, then knotted . If obstruction, infection or pain → unknot and it drains as a nephrostomy
None	—	—	Requires flawless technique

- A Foley catheter for 24–48 h is **dogma without hard evidence**; it makes sense only with a double-J.

Outcome and Complications

- Pyeloplasty is safe, with **90–100% success**. However, “**success**” is defined **inconsistently** (symptom relief, stable or improved function, falling AP diameter, or, most rigidly, no reintervention), which makes studies hard to compare.
- **Bleeding** is controlled with bipolar diathermy and is the **commonest reason for conversion**, followed by lack of progress and difficult UPJ dissection.
- **Early failure signs:** infection, pain and increasing dilatation. These are often due to **clot or oedema, which settle in days**, sometimes with temporary nephrostomy.

- **Late failure** (falling function, increasing dilatation) → **balloon dilatation or incision with stenting for 4–6 weeks**, or **redo pyeloplasty, preferably transperitoneal** because the retroperitoneum is scarred.
- **Severe ureteric scarring, too short for tension-free repair**, plus a **dilated lower-pole calyx with thin parenchyma** → **uretero-calicostomy** (open, laparoscopic or robotic).

Figure legend, Fig. 53.17 (uretero-calicostomy) — Mobilise the kidney → **lower-pole nephrectomy** to expose a dilated calyx widely → spatulate the ureter laterally → first suture at the spatulation apex to the lateral calyx and a second at 180° → place all sutures before tying, over an **internal stent ± nephrostomy** → close the capsule over the cut surface but **not near the anastomosis** (extrinsic compression) → protect with **perinephric fat, peritoneal or omental flap**.

KEY POINTS: PYELOPLASTY

- UPJO is **intrinsic narrowing** or **extrinsic obstruction by a lower-pole vessel** passing anterior to the UPJ or proximal ureter.
- Commonest associated anomaly: **contralateral UPJO**, then renal dysplasia and multicystic dysplastic kidney.
- **Dismembered pyeloplasty** is the favoured repair.
- Laparoscopy gives excellent views and cosmesis and **matches open results** in the short term; **robotic assistance eases the anastomosis**.
- Complications: prolonged drainage, failure of function or hydronephrosis to improve → endoscopic treatment or, **more often, redo dismembered pyeloplasty**.

Concept — Repair of UPJO has **one fixed principle and many variable routes**. Dismember, excise, spatulate and anastomose without tension. Access, stent and robot are choices of convenience around that constant, and failure is managed by repeating the same principle.

Ureteral Strictures

- **Congenital ureteral strictures are rare**. They are detected antenatally as dilatation and **usually misdiagnosed as hydronephrosis or megaureter**.
- Postnatal ultrasound, MRI and retrograde pyelography confirm a **mid-ureteric stricture**.
- Treatment: **excision and ureteroureterostomy**. Excised segments are **1–3 cm** long and show **increased collagen and muscular hypertrophy**.

Ureteral Polyps

- Rare. Usually **upper third of the ureter** → **UPJO**. **Male and left-sided predominance**; single or multiple; up to several centimetres.
- Slow-growing and benign, so they usually present **later in childhood with pain and haematuria** rather than antenatally.
- Imaging: ultrasound, contrast CT, MR urography. **Confirmed by retrograde pyelography or ureteroscopy**.
- Treatment: **endoscopic polypectomy**; **dismembered pyeloplasty with resection** when near the UPJ; or **segmental resection with ureteroureterostomy**.
- Histology: **benign fibroepithelial polyp**.

Retroperitoneal Fibrosis in Children

- **Extremely rare** in children. **Mostly idiopathic**; also autoimmune, infective or neoplastic.
- Signs: **falling renal function, bilateral hydronephrosis, hypertension**. Imaging: ultrasound and MRI.

- Treatment escalates: **pulsed steroids** → **azathioprine** → **double-J** → **ureterolysis**.

VESICoureTERIC REFLUX

Definition and Clinical Presentation

- **VUR is the abnormal retrograde flow of urine from the bladder to the upper tract.** When symptomatic, it causes **UTI, pyelonephritic scarring and, at worst, end-stage renal disease.**
- High-grade VUR is associated with **voiding dysfunction**; which causes which is unresolved.
- **The aim of surgery is one-way, unobstructed, antegrade flow**, achieved by reimplanting the ureter in an antirefluxing fashion. There are open, laparoscopic and endoscopic options.

Surgical Management

KEY POINTS: SURGICAL PRINCIPLES OF VUR CORRECTION

- **Exclude or treat secondary causes of reflux** first.
- **Handle the ureter gently** to preserve its blood supply.
- **Handle the bladder gently** to limit postoperative haematuria and spasm.
- **Mobilise enough ureter** for a tension-free anastomosis.
- **Tunnel length-to-diameter ratio 5:1** (Paquin).
- **Keep the course straight**, allowing for bladder distension, which can angulate or kink the ureter at either extreme of filling.

- **Classification of procedures:** by access (**open or laparoscopic**), approach (**intravesical or extravesical**) and tunnel position relative to the original hiatus (**suprahiatal or infrahiatal**).
- **Open set-up:** supine, hips slightly abducted. Prepare abdomen, perineum and genitalia so the urethra can be catheterised in the sterile field. **Prophylactic intravenous (IV) antibiotics** in most cases.

Incision

- **Pfannenstiel incision** in a skin crease **2 cm above the symphysis**, to the lateral edges of the recti → anterior rectus sheath opened transversely and raised from symphysis to just below the umbilicus → recti separated in the midline along the linea alba → perivesical space.
- **Vertical lower midline incision** for **redo surgery**, because it allows more cranial ureteric mobilisation.
- A **moderately filled bladder** is easier to identify and mobilise.

Intravesical Procedures

Common mobilisation sequence: sweep peritoneum off the dome → open the bladder vertically between stay sutures to **within 2 cm of the bladder neck** → pack the dome with saline gauze → **Denis Browne ring retractor** (blades on the rectus bellies plus a cranial blade countering the packed dome); lateral walls held by stay sutures, not blades, to spare friable mucosa → **cannulate the orifice with a 5-Fr feeding tube**, fixed with 5-0 monofilament as a traction handle → **circumscribe the orifice 1-2 mm away** with fine low-wattage needle diathermy → divide intra- and extravesical attachments with tenotomy scissors and bipolar diathermy, **staying in the correct plane to preserve distal blood supply** → Langenbeck retractor at 12 o'clock and peritoneum swept off with a pledget → ureter **reaches the contralateral wall without tension** → choose the tunnel.

Politano–Leadbetter Technique

- Politano and Leadbetter, **1958**. **Suprahiatal**: a **new hiatus superomedially** with a vertical tunnel running straight down to the native hiatus. Suited to **high-grade VUR needing a long tunnel**.

Sequence: right-angled clamp through the original hiatus towards the planned neo-hiatus, sweeping peritoneum off the posterior bladder → push through, incise and widen. **Danger**: **bowel, vas deferens, vagina**, so retract the superior lip of the old hiatus and sweep under direct vision → 3-0 guide suture extravesically from the new to the old hiatus → ureter drawn out and back in through the neo-hiatus → **close the old detrusor defect (4-0)** → submucosal tunnel **capacious enough not to constrict** → shorten ± spatulate → **hitch to detrusor at 6 o'clock (5-0)** → mucosa to mucosa with **6-0 interrupted** → close the neo-hiatus mucosal defect (5-0) → bladder closure (4-0) and **Foley for 1–3 days; two-layer closure shortens catheter time**. Stents only in complex cases.

Glenn–Anderson Technique

- Glenn and Anderson, **1967**. **Infrahiatal**: tunnel advanced **distally towards the bladder neck**.
- Advantage: **least liable to angulation and kinking**, so obstruction is avoided. Limitation: **tunnel length is capped by the distance to the bladder neck**. Success up to 98%.

Cohen Cross-Trigonal Technique

- Cohen, **1977**. **Transhiatal, cross-trigonal** tunnel towards the opposite orifice, giving a **longer tunnel**. The **commonest intravesical procedure**: versatile, simple and reliable, with **success over 98%**.
- **Suited to thick-walled trabeculated bladders** (trigone undistorted) and to **concomitant bladder-neck reconstruction** (transverse tunnels leave length for the neck).
- **Main drawback**: **difficult later retrograde catheterisation** for imaging, stents or stones. Solutions are a **flexible ureteroscope**, or **cystoscopy plus a suprapubic trocar or IV cannula** aimed at the orifice to pass a guidewire.
- **Unilateral**: new orifice **just above the contralateral orifice**. **Bilateral**: the **more laterally displaced (superior) ureter** goes above the contralateral orifice, and the second ureter goes to the **inferior edge of the first ureter's native hiatus**. A **common tunnel** also gives excellent results.

Extravesical Procedures

- **Lich–Gregoir**: Lich (1961) and Gregoir (1964) independently. The **bladder is not opened**, so **no postoperative haematuria or bladder spasm**. The detrusor is incised down to intact urothelium and the ureter laid in the detrusorotomy to create the tunnel.
- **Main concern**: **transient postoperative voiding dysfunction and retention in up to 10%**, needing catheterisation, **especially after bilateral procedures**. The cause is **injury to pelvic-plexus innervation** of the detrusor and trigone.
- **Neuroanatomy**: cadaveric work shows the **main pelvic plexus lies about 1.5–2 cm dorsal and medial to the UVJ**, with small branches running along the **medial side of the ureter outside the mesoureter**. Fetal immunohistochemistry with 3D imaging places the nerves on the **medial border just outside Waldeyer's sheath**, extending circumferentially and **distal and lateral to the UVJ** along the detrusor.
- **Practical rule**: **dissect hugging the ureter, avoid the dorsomedial UVJ region, and stop at the UVJ**. This return to the original description has reduced voiding dysfunction.

Open Lich–Gregoir sequence: Pfannenstiel, or **inguinal incision for unilateral cases** → bladder **one-third full**, peritoneum cleared → **divide the obliterated umbilical artery** → peritoneum teased off the anterior ureter, staying close (nerves) while minding its blood supply → vessel loop as handle → **mobilise only to the UVJ** → with the bladder moderately distended in its anatomical position, **mark a 5-cm tunnel** along the ureter's

natural course → retract the bladder medially → low-wattage monopolar detrusorotomy; assistant lifts fibres → **uniform mucosal bulge** → raise detrusor flaps bluntly, wide enough not to constrict → apex stay suture controls length → close the detrusor **caudocranially with interrupted 4-0, catching ureteric adventitia, especially at the apex**, to stop slippage or prolapse → **Foley for 12–48 h**; discharge once voiding.

- **Zaontz modification**: detrusorotomy carried **around the UVJ** for advancement sutures into the trigone. This encroaches on the nerves and probably caused much of the voiding dysfunction of that era, so it is **now avoided**, including laparoscopically.

URETERIC REIMPLANTATION TECHNIQUES COMPARED

Technique	Approach	Tunnel	Strength	Weakness
Politano–Leadbetter (1958)	Intravesical (combined)	Suprahiatal , vertical from a new superomedial hiatus	Long tunnel for high-grade VUR	Blind neo-hiatus (risk to bowel, vas, vagina); kinking if neo-hiatus too lateral or anterior
Glenn–Anderson (1967)	Intravesical	Infrahiatal , towards the bladder neck	Least kinking	Short tunnel (limited by bladder neck)
Cohen (1977)	Intravesical	Cross-trigonal, transhiatal	Commonest ; simple; trabeculated bladder; bladder-neck reconstruction	Difficult retrograde access later
Lich–Gregoir (1961/1964)	Extravesical	Detrusorotomy along the natural course, 5 cm	No haematuria or spasm ; short catheter time	Voiding dysfunction, especially bilateral
RALUR-EV	Robotic extravesical	Y-shaped detrusorotomy, 4–5 cm	Less pain, spasm and stay	Early series: leak, obstruction, injury

Concept — Intravesical repairs buy **tunnel length at the cost of bladder morbidity**, while the extravesical repair spares the bladder lumen at the cost of **pelvic-nerve risk**. Every refinement of the Lich–Gregoir operation is a lesson in where the nerves run.

Minimally Invasive Procedures

- Laparoscopic extravesical reimplantation was shown feasible in the mid-1990s. It brought **less pain, haematuria and spasm and a shorter stay**, and led on to robot-assisted (da Vinci) surgery.
- **AUA Paediatric Vesicoureteral Reflux Guidelines Panel meta-analysis (Elder, 1997)**: open reimplantation success **95.9%** (86 studies; 6,472 patients; 8,563 ureters). This is the open benchmark.
- **Early minimally invasive series fell short of it**, with lower reflux resolution and **complications unusual after open surgery: ureteric injury, obstruction and leak**, especially after **intravesical vesicoscopic** repair.
- Caveat: **minimally invasive series had stricter radiographic follow-up** than historical open series. With experience, results are **converging on open outcomes**.

Robot-Assisted Laparoscopic Ureteral Reimplantation — Extravesical (RALUR-EV)

Sequence: **low lithotomy**, prepare from xiphoid to perineum, catheter in the field for drainage and filling → **12-mm camera port by Hasson technique at the umbilicus** → **two 8-mm robotic ports** in the midclavicular line just below the umbilicus, plus an optional 5-mm assistant port → **dock between the legs** (right-handed surgeon: bipolar left, monopolar scissors right) → identify the ureter at the **pelvic brim** → mobilise to the UVJ, **protecting the vas deferens or uterine artery**, keeping close to the ureter to **avoid the dorsomedial neurovascular bundle** → bladder stay stitch for traction → **partially fill the bladder** → **4–5-cm Y-shaped**

detrusorotomy along the ureter's course → raise flaps → lay the ureter in → **running 4-0 absorbable monofilament, caudocranial, catching adventitia (especially at the apex)** to prevent slippage.

- The figure legend adds that an umbilical tape round the ureter gives atraumatic handling, and an apex stay suture maintains orientation and tunnel length.

Postoperative Evaluation

- In expert hands, ureteroneocystostomy for **low-grade primary VUR approaches 100% success**.
- **Ultrasound at 6–12 weeks** postoperatively for all patients (AUA VUR Guideline, 2010, amended 2017).
- One-year resolution is **significantly higher for low-grade than high-grade** reflux.
- **Voiding cystourethrogram (VCUG) can be omitted** after low-grade primary VUR when pre- and postoperative ultrasound are normal and there is **no voiding dysfunction or recurrent UTI**.
- **Postoperative VCUG is still reasonable** with a **dysfunctional bladder, new hydronephrosis or UTI**, or after **high-grade reflux**, where success is lower.

Complications of Ureteral Reimplantation

Early Complications

- **Persistent reflux** is usually clinically unimportant and **commonly resolves by 1 year**. Persistence at 1 year is concentrated in children who had **high-grade reflux** preoperatively. Early low-grade reflux is attributed to **postoperative bladder inflammation and dysfunction**, which settle.
- **Contralateral reflux** after unilateral reimplantation occurs in a **small minority (roughly one in ten)**, and **most resolve spontaneously**. Routine bilateral reimplantation is therefore **not justified**.
- **Risk factors for contralateral VUR**: age <6 years, bladder capacity <50% of expected, reimplanted **refluxing duplex system**, and a trend with higher corrected grade. **The surgical technique used makes no difference**.
- Proposed mechanisms (speculative): distortion of the contralateral hemitrigone; **pop-off effect**; **bilateral reflux missed on the initial cystogram**; intermittent natural history.
- Management: **observation in most**; **prophylactic antibiotics in asymptomatic children under 4–5 years**; intervention for clinical pyelonephritis. **Repeat VCUG is unnecessary if infection-free**.
- **Obstruction: mild–moderate early hydronephrosis is common and resolves**, after both open and robotic extravesical repair. **Rule of thumb: postoperative dilatation should not exceed the maximal dilatation the reflux produced**.
- **Acute obstruction** results from a twisted or kinked ureter, intramural clot, submucosal haematoma or oedema. It presents **within the first 2 weeks** with pain, nausea and vomiting, and infection here is serious. Ultrasound, then **delayed excretion on nuclear scan**, confirm it. Drain with a **retrograde double-J or percutaneous nephrostomy**, and **internalise the nephrostomy early to avoid a “dry reimplant”**. Many resolve without further surgery.

Long-Term Complications

- **Obstruction** has several causes. **Distal ureteric ischaemia** follows aggressive dissection or **high pressure in a non-compliant bladder** (neurogenic, posterior urethral valves). **Angulation** follows a **neo-hiatus placed too lateral or anterior**, with kinks that **worsen as the bladder fills**. Other causes are a **constricting, underdeveloped tunnel** and **neo-orifice stenosis**.
- Management of obstruction escalates: **balloon dilatation and stenting**, or **unroofing the distal tunnelled ureter** while trying to keep the antireflux mechanism → **re-reimplantation** if these fail.

- **Recurrent or persistent reflux: failure in primary low-grade reflux is extremely rare.** Most failures come from **high-grade reflux or an inadequate tunnel-length-to-diameter ratio** (short tunnel, untapered wide ureter).
- Another major cause is **unrecognised secondary reflux** (neurogenic or posterior urethral valve bladder). **Optimise storage and emptying first with anticholinergics plus clean intermittent self-catheterisation**, which usually resolves the reflux.
- **Reimplanting into an abnormal bladder** only worsens dilatation and scars the pelvis, which makes later repair harder.

	Early	Long-term
Reflux	Ipsilateral persistence usually resolves by 1 year; contralateral reflux mostly resolves	Rare after low-grade disease; due to high grade, a short tunnel or secondary bladder pathology
Obstruction	Oedema, clot, haematoma or kink ; within 2 weeks; drain and wait	Ischaemia, angulation, tight tunnel or stenosis ; dilate, stent, unroof or redo

Concept — Early complications are **transient tissue reactions** that are drained and waited out. Late ones are **errors of design** (blood supply, hiatus position, tunnel ratio, the unaddressed bladder) that need correction. **A reimplant fails when the bladder, not the ureter, was the problem.**

Reoperative Reimplantation

- **Combined extravesical and intravesical mobilisation; excise ischaemic segments** until there is **free bleeding and peristalsis**; create a **new hiatus and tunnel**.
- **Short ureter → psoas hitch**, with non-absorbable sutures, **placed before the tunnel is made**. The bladder is fixed to the psoas sheath either side of the iliac vessels, and in children can reach almost to the **common iliac bifurcation**.
- **Psoas hitch on one side only**: bilateral hitches give inadequate length on both. **Both ureters short → hitch one side plus transureteroureterostomy** for the other.
- **Boari flap**: a flap from the dome to the anterior wall, **based on the posterior wall** and rotated up, wide enough to tunnel and tubularise.
- **Nipple valve** plus a short tunnel: made by **spatulating the ureter and folding it back on itself**; especially useful for **dilated ureters**.
- **Severely foreshortened ureter → reconfigured bowel ureter** (colon or ileum, **Monti-type**). A long tube comes from a short segment **without tapering**, so there are **no metabolic consequences**, and the **central mesentery eases tunnelling**. This is superior to the classic tapered ileal ureter.

ENDOSCOPIC TREATMENT OF VESICoureTERAL REFLUX

- Matouschek introduced endoscopic injection in 1981 using Teflon. O'Donnell and Puri popularised it as **STING** (subureteric Teflon injection), 1986.
- Teflon was never approved by the US Food and Drug Administration (FDA), because of **migration of polytetrafluoroethylene (PTFE) particles**.
- **Dextranomer/hyaluronic acid copolymer (Dx/HA; Deflux)** is now the most used agent. Initial resolution of **low-grade primary VUR after one injection is about 90%**, but later results are less consistent because technique and definitions of success vary.

Technique of Endoscopic Injection

Sequence: antibiotic at induction → cystoscopy before opening the implant material (abandon if the bladder is inflamed) → offset-lens injection cystoscope for a rigid needle; needle 3.7–5 Fr by material viscosity; viscosity also decides between a **plain syringe and a ratcheted holder** → optional **3-Fr ureteric catheter** to lift the roof and show the tunnel axis → **needle bevel up at 6 o'clock** → inject slowly: **a mound should appear within the first 0.1–0.2 mL**; if not, the tip is misplaced → reposition by slow withdrawal while injecting and rotating the bevel → aim for a **“volcano” with the orifice on top** → continue until the orifice is **crested- or slit-shaped** → hold the needle **1 minute** to limit extrusion (**not essential with Deflux**) → **inspect the mound with the bladder empty and full**.

INJECTION TECHNIQUES COMPARED

Technique	Needle entry	Material placed	Result
STING (classic)	2–3 mm distal to the orifice , advanced 4–5 mm submucosally	Under bladder mucosa just below the orifice	Baseline
HIT (hydrodistension implantation technique)	Inside the hydrodistended orifice	Submucosally within the intramural ureter	Better than STING for all grades
Double HIT	Two intraureteric punctures (proximal and distal)	Intramural ureter, two depots	Further improvement , durable at intermediate and long term

- **Accurate needle entry is paramount.** A wrong puncture fails to form a supporting mound, and repositioning lets material leak out of the first puncture.
- **Mound shape predicts success: a volcano mound does markedly better** than other shapes.
- **Puncture bleeding** → empty the bladder and press with the scope tip. **Do not cauterise:** mucosa sloughs and the implant extrudes.
- Afterwards: urethral lidocaine gel, **no catheter**, brief recovery, **day case, immediate return to activity**.
- **There is a significant learning curve.**

Figure legends, Figs 53.24–53.25 — Sites 1 and 2 are the double HIT (intramural, transmural ureter); site 3 is the classic STING (just below the orifice). After HIT, the mound shows a **slit-like orifice**.

Follow-Up

- **Antibiotics for 3 months. Ultrasound at 2–6 weeks** to exclude acute obstruction by the bulking agent. **Ultrasound ± VCUG at 3 months**, depending on the clinical picture; if the child is clinically well, ultrasound alone suffices.
- **Persistent VUR** → 1–2 repeat injections → then open, laparoscopic or robotic reimplantation.
- **Late-onset ureteric obstruction** after injection is increasingly reported, which argues for **long-term ultrasound follow-up**.
- **Open surgery after Deflux is usually not more difficult.** The implant is absent or **well encapsulated in the wrong plane** and comes out **en bloc**. Other agents have caused more difficulty.

Materials Used for Endoscopic Correction of Reflux

- **Ideal agent:** non-toxic, stable, **non-migrating**; minimal inflammation but **well encapsulated by fibrous tissue**; injectable through a long fine needle; **viscous enough not to leak**; keeps its volume and shape after the carrier is excreted.
- **Particulate agents carry a migration risk; degradable agents carry a durability problem.**

	Intravascular migration	Phagocytic migration
Mechanism	Bolus expansion ruptures small vessels → particles enter the circulation	Uptake by tissue macrophages or blood monocytes
Size threshold	<50 µm passes the pulmonary bed into the systemic circulation	<80 µm can be phagocytosed

BULKING AGENTS: COMPOSITION AND PROPERTIES

Agent	Composition	Particle size	Biodegradable	FDA status
Teflon (PTFE paste)	Polytetrafluoroethylene paste	90% <40 µm	No	Never approved
Macroplastique	Polydimethylsiloxane, a solid silicone elastomer	—	No	—
Deflux (Dx/HA)	Cross-linked dextranomer microspheres in stabilised sodium hyaluronate gel	80–250 µm	Yes: gel reabsorbed; microspheres encapsulated by fibroblasts and collagen	Approved 2001
Vantris (polyacrylate polyalcohol copolymer, PPC)	Acrylic particles in glycerol and saline carrier	300 µm	No: fibrotic encapsulation, little inflammation	Not approved

BULKING AGENTS: SIGNATURE CONCERN AND OUTCOME

Agent	Signature concern	Outcome and status
Teflon	Particle migration	Durable (low late recurrence) but all but abandoned
Macroplastique	—	Seldom used now
Deflux	Volume loss (about 23% beyond 3 months); durability unproven ; late-onset obstruction reported	Preferred agent in most centres: natural product, no ratcheted syringe, finer needle . Single-injection success varies by series. Double HIT gives about 90% resolution . Long-term follow-up after STING: most maintained grade ≤1. Salvage reimplant is easy (implant periureteric and encapsulated)
Vantris	Late-onset ureteric obstruction, a major concern	Higher resolution than Dx/HA , even for high grade; follow-up up to 4.5 years

- Other autologous and non-autologous agents are not yet in clinical use.

Concept — Bulking agents trade **migration against durability through particle size and degradability**. Small permanent particles migrate (Teflon); large degradable ones shrink (Deflux); large permanent ones stay and work but can obstruct late (Vantris).

Recurrence of Vesicoureteral Reflux After Endoscopic Correction

- **The true incidence is unknown**, because VCUG is rarely repeated after an initial negative study outside research protocols.
- **Teflon: low long-term recurrence** (follow-up up to 17 years). **Swedish Reflux Trial: 20% recurrence at 2 years** after initially successful Dx/HA. Higher recurrence with an **absorbable agent is expected**.
- Without UTI, **proving long-term absence of VUR matters less**. Even if the agent resorbs, **growth and UVJ maturation may maintain resolution**, so the injection may simply have “**bought time**.”
- **Higher risk of recurrent UTI and late VUR** despite a negative VCUG: **multiple febrile UTIs before injection, bladder and bowel dysfunction, renal scarring**.

- Endoscopic correction is a **reasonable alternative** to surgery, but families must be told that **success varies between centres and may not be durable**. The significance of recurrence in young adulthood is undefined.

MEGAURETER

Definition

- **Megaureter is descriptive: a dilated ureter, whatever the cause.** Coined as “megaloureter” by Caulk (1923); synonyms are wide ureter and hydroureter.
- **Megaureter = diameter ≥ 7 mm.** The upper limit of normal ureteric diameter up to 16 years is 5.0–6.5 mm (Cussen; Hellström).
- **Primary** (intrinsic to the ureter) versus **secondary** (bladder pathology: neurogenic bladder, outlet obstruction, infection).
- **Four categories by cause: obstructed • refluxing • non-obstructed non-refluxing • refluxing with obstruction.**
- Refluxing megaureter is easily diagnosed. **The difficulty is telling obstructed from non-obstructed non-refluxing megaureter**, because dilatation, however alarming, does not prove obstruction.
- **The challenge is balance:** catch obstructed ureters before function is lost, without operating on stable, balanced dilatation.

Etiology, Occurrence, and Presentation

- **Primary obstructive megaureter (POM):** an **adynamic terminal segment at or near the UVJ** → absent or inefficient peristalsis → functional obstruction and dilatation upstream.
- **Developmental basis:** ureteric muscle **matures craniocaudally**, so the **juxtavesical segment matures last, continuing after birth**. Fetal circular fibres become the double layer of the term infant only well after delivery, which explains **transient UVJ dysfunction and spontaneous resolution with age**.
- Other findings: **excess abnormal collagen** at the narrow segment; a **thick enveloping muscle sleeve**; and abnormal **interstitial cells of Cajal** (pacemaker cells for peristalsis).
- **Incidence:** 10–23% of antenatally detected upper-tract dilatations. Young males, left side; bilateral in 25%; contralateral dysplasia or obstruction in 10–15%.
- **Most are detected antenatally and are asymptomatic.** Before routine ultrasound, POM was a rare diagnosis among symptomatic dilatations; antenatal screening now detects the majority.
- Symptoms: **UTI, abdominal pain, microscopic haematuria** (even without infection, from distension or calculi). Rarely, an **abdominal mass or renal impairment**, as in the pre-ultrasound era.

Surgical Indications

- **Most non-refluxing megaureters are benign and resolve in the first few years.** Complete resolution and its speed are **inversely related to ureteric diameter**.

	Ureteric diameter <10 mm	Ureteric diameter ≥ 10 mm
Course on prophylaxis	Minimal complications or febrile episodes	Recurrent febrile UTI, stones, abdominal pain
Resolution	Usual	Only a minority resolve completely

	Ureteric diameter <10 mm	Ureteric diameter ≥10 mm
Surgery	Virtually none	A substantial minority

- **Persistent dilatation** → **periodic ultrasound plus functional studies** until resolution or stability. **Extend follow-up into early adulthood**, because some stable megaureters worsen at or after puberty.
- For 25 years the trend has been **towards conservative and then minimally invasive management**.
- **Operate when there are: symptoms with recurrent UTI · progressive unremitting dilatation · DRF <40% · a fall in DRF of ≥5% on sequential, comparable nuclear studies.**
- **Do not rely on diuretic renogram washout or T½.** Neonatal curves are affected by renal function and diuretic response, hydration, posture, collecting-system volume and compliance, diuretic timing and interpretation. **False-positive rates reach 44%.**

Surgical Management

- Reserved for the **experienced paediatric urologist**.
- **Principle:** excise the stenotic distal segment → **straighten** → **taper** → **non-refluxing reimplant at a 5:1 length-to-diameter ratio**. The ratio improves luminal coaptation, giving effective peristalsis and urine transport. It is increasingly done laparoscopically or robotically.

TEMPORISING OPTIONS IN THE INFANT WITH POM

Option	How	Pros	Cons
Nephrostomy	Percutaneous	Prompt decompression	Hard to maintain in infants ; leak, infection, displacement
Distal cutaneous ureterostomy	Oblique inguinal incision → muscle-split → perivesical space → ureter out as a loop, fixed to fascia (5-0) → opened transversely, edges everted to skin (5-0 or 6-0) → drain 24–48 h	Rapid decompression of an infected system; delays reconstruction; stomal stenosis rare (wide ureter)	Needs later reconstruction
Refluxing reimplant (internal diversion)	Transect above the stenosis; re-anastomose freely refluxing	Converts “dangerous” obstruction into the “lesser evil” of reflux ; promising early results	Long-term effect of reflux on renal function awaited
Endoscopic dilatation and stenting	See below	Least invasive	Complications in up to a third

- **Why delay definitive repair in infants:** the **small bladder and the massive ureter are incongruent**, so a safe non-refluxing tunnel is impossible. Fear of iatrogenic bladder dysfunction **may be overstated**.
- **Definitive repair:** Pfannenstiel → mobilise as for intravesical reimplant → continue **intravesically (Cohen, Politano–Leadbetter)**, or **work extravesically for a very large ureter** with a large bladder-ureter mismatch, to avoid excessive bladder dissection and a **huge trigonal hiatus**.

URETERIC TAILORING: PLICATION VERSUS EXCISIONAL TAPERING

	Starr plication	Kalincinski folding	Hendren excisional tapering
Principle	Imbricate the redundant lateral wall inward	Exclude the lateral lumen and fold it back over the functional lumen	Excise redundant lateral ureter

	Starr plication	Kalicinski folding	Hendren excisional tapering
Technique	Over an 8–10-Fr catheter; Allis clamp marks; interrupted 5-0 Lembert sutures	Continuous 6-0 woven craniocaudally creates two lumens; folded and tacked with interrupted 6-0	Over an 8-Fr (infant) or 10-Fr catheter; running locking 6-0 on the proximal two-thirds, interrupted on the distal third so it can be shortened at reimplantation
Blood supply	Preserved (walls intact) → less ischaemia and stenosis		At risk → ischaemic stricture
Limit	Only moderate dilatation (<1.75 cm) ; a bigger ureter becomes too bulky to reimplant, with more complications and failures		Massively dilated ureters unsuitable for folding
Success	90–95%		>90%

- **Hendren's excisional tapering (1969) came first**; folding techniques were developed because of its **vascular risk**. A refinement excises only **mucosa and muscle**, leaving the **vascular adventitia**, to lower ischaemia further.
- **Taper gradually**: an abrupt change in calibre is itself obstructive.
- Stents after tailoring: **feeding tube for 7 days** or **double-J for 4 weeks**. **No stentogram is needed at removal**. Then reimplant on VUR principles.
- **Laparoscopic or robotic** remodelling and reimplantation gives **short- to medium-term results comparable to open**. **Extravesical repair dominates** because tapering inside the bladder is technically limited, although **pneumovesical excisional tapering with transtrigonal reimplant** is feasible.

Outcomes

- Remodelling and reimplantation for POM is **durable, with over 90% success**.
- **POM does better than refluxing megaureter**, attributed to a **higher collagen-to-smooth-muscle ratio** at the refluxing UVJ, which gives a **stiffer, less compressible ureter**.
- **Worse outcomes** with dysfunctional voiding, neurogenic bladder, posterior urethral valves or outflow obstruction.
- **Major complications: obstruction, VUR, persistent dilatation**.
- **Obstruction** → **stent first** (may be oedema). **Ischaemic stricture**, especially after excisional tapering → **redo ureteroneocystostomy**.
- **Reflux** → **conservative management**, especially low grade, because it often resolves. **Subureteric injection** works with few complications. If injection fails or febrile UTIs recur → revision surgery.
- **Transureteroureterostomy** is an option for **unilateral** cases with **severe scarring or doubtful blood supply**.

Dilation and Stenting

Sequence: endoscopic access to the POM orifice → **dilate with a ureteric catheter or high-pressure balloon under fluoroscopy until the waist disappears** → **double-J for 2–6 months** → persistent dilatation or falling function → **re-stent**, or **reimplant**.

- **Short- to medium-term success about 70–80%**, but the **evidence is weak**: mostly retrospective, with heterogeneous indications and definitions, and **follow-up rarely beyond 5 years**.
- **Complications in up to a third**: stent migration, infection, haematuria, VUR, stones. There may also be **re-stenting, open stent placement or ureteric injury needing emergency reimplantation**.

KEY POINTS: MEGAURETER

- **Megaureter is descriptive:** a dilated ureter of any cause.
- **Four categories:** obstructed, refluxing, non-obstructed non-refluxing, refluxing with obstruction.
- The challenge is to **prevent loss of renal function**, whether by conservative or surgical treatment.
- **Surgery for:** recurrent UTI, progressive dilatation, **DRF <40%, DRF fall $\geq 5\%$** on comparable sequential scans.
- **Excise the stenotic segment → straighten → taper → non-refluxing reimplant at 5:1.**
- **Most surgeons use temporary postoperative stenting.**

Concept — Primary obstructive megaureter is **the UPJO problem moved to the bottom of the ureter**: an immature adynamic segment that usually matures on its own. Surgery is kept for the ureter that is too wide or losing function, and **its size dictates the method**: divert or leave refluxing in infancy, fold the moderate ureter, excise the massive one.